INCH-POUND
MIL-DTL-62366D(AT)
07 July 2010
SUPERSEDING
MIL-DTL-62366C(AT)
30 September 1998

DETAIL SPECIFICATION

RING ASSEMBLIES, ELECTRICAL CONTACT

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification defines the requirements for two electrical contact ring assemblies, referred to herein as "slip ring assemblies", for use on military ground vehicles (see 6.1).

1.2 Classification. The slip ring assemblies covered by this specification will be of the following classes, as specified in table I (see 6.2 and 6.5).

TABLE I. Classification of slip ring assemblies.

Class	Number of rings			
	Control			
	Power	Signal	Data	Shield
I	2	15		
II	6	11	16	5

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all

Comments, suggestions, or questions on this document should be addressed to U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRANS MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to DAMI_STANDARDIZATION@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil/online/.

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specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

FEDERAL STANDARDS

FED-STD-595/24533 - Seafoam Green, Navy

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts, or Equivalent (See 4.3.1)
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines, or Equivalent (See 4.3.1)
MIL-STD-1275	-	Characteristics of 28 Volt DC Electrical Systems in Military Vehicles

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-6083	-	Hydraulic Fluid, Petroleum Base, for Preservation and Operation
MIL-PRF-22750	-	Coating, Epoxy, High Solids
MIL-PRF-23827	-	Grease, Aircraft and Instrument, Gear and Actuator Screw
MIL-PRF-46170	-	Hydraulic Fluid, Rust Inhibited, Fire-Resistant, Synthetic Hydrocarbon Base

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094 or https://assist.daps.dla.mil/quicksearch/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract (see 6.2).

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ARMY DRAWINGS

12265512	-	Ring, Electrical Contact (M901)
12310155	-	Slip Ring Assembly, Electrical Contact (M981)

(Copies of these documents are available from dami_standardization@conus.army.mil or U.S. Army RDECOM, Tank Automotive Research Development, and Engineering Center, ATTN RDTA-EN/STND/TRANS, MS #268, 6501 E 11 Mile Road, Warren, MI 48397-5000)

2.3 Non-Government documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract. (see 6.2).

NATIONAL AEROSPACE STANDARDS COMMITTEE (NA/NAS)

NASM90726	-	Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Steel, Grade 5, Cadmium Plated, UNF-2A

(Copies of this document may be purchased from the Aerospace Industries Association of America 1000 Wilson Boulevard, Suite 1700, Arlington, VA 22209-3928. or website http://www.aia-aerospace.org.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA-HP4	-	Electrical and Electronic FEP (Fluorinated Ethylene Propylene) Insulated High Temperature Hook-Up Wire, Types KT (250 V), K (600 V), and KK (1000 V).
NEMA-WC-27500	-	Standard for Aerospace and Industrial Electrical Cable

(Copies of NEMA documents are available from www.nema.org or National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1752, Rosslyn, VA 22209.)

SAE INTERNATIONAL

SAE AS22759/7	-	Wire, Electric, Fluoropolymer-Insulated, Abrasion Resistant Extruded PTFE, Medium Weight, Silver-Coated Conductor, 600 Volt
SAE AS81044/6	-	Wire, Electric, Crosslinked Polyalkaline Insulated, Tin-Coated Copper, Normal Weight, 600-Volt, 150 Degrees C

(Copies of SAE International documents are available from www.sae.org or SAE Customer Service, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

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2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2 and 6.3), a sample shall be subjected to first article inspection in accordance with 4.1.1.

3.2 Design and construction. The slip ring assembly shall consist of a housing and mounting base assembly, a brush block and terminal assembly, and a rotor assembly, in accordance with the following drawings and 3.2.1 through 3.2.3 (see 4.3.3):

Class I - Class I slip ring assemblies shall be in accordance with Drawing 12265512
Class II - Class II slip ring assemblies shall be in accordance with Drawing 12310155

3.2.1 Housing and mounting base assembly. A housing and mounting base assembly with seals shall provide a splash proof enclosure and suitable mounting provisions (see 4.3.3.1).

3.2.2 Brush block and terminal assembly. A brush block and terminal assembly shall consist of two spring loaded brushes per ring with barriers that will positively retain the brushes in the conducting area (see 4.3.3).

3.2.3 Rotor assembly. The rotor assembly shall be internally and mechanically retained to prevent axial motion relative to the housing. The number, type, and performance characteristics of the rings shall be as specified in 3.4.6 (see 4.3.3).

3.2.4 Interface requirements. The hookup wires shall be as specified in tables II and III (see 4.3.3.2).

TABLE II. Class I hookup wire requirements.

Pin	Wire type and size	Alternate wire type and size
A, B, D, E, F, G	NEMA-WC27500-16RC2S05	NEMA-WC27500-16ME2T08
H through M, U, V	NEMA-WC27500-16RC4S05	NEMA-WC27500-16ME4T08
P, T, R	SAE-AS22759/7-6	NEMA-HP4 or SAE-AS81044/6-8
W, X	SAE-AS22759/7-6	MIL-W-16878/12-KK-4 or MIL-W-81044/6-12
C, O	Shield return	Shield return

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TABLE III. Class II hook up wire requirements.

Connector	Pin	Wire type and size
J1	A, D B, C, E, F	2 - # 6 Braided SAE-AS81044/6-8-9
J2	A through F, J through M, P, R through Z, a, b, c, e, f, g, h H G, N, W, d, j	NEMA-WC27500-20ML2T08 NEMA-WC27500-20ML1T08 Shield return

3.3 Materials. Materials used shall be in accordance with applicable drawings (see 3.2 and 3.3.1 through 3.3.6). When not thereby designated, the materials shall be in accordance with the manufacturer's materials specifications for slip ring assemblies and shall meet all of the operating and environmental requirements as specified herein (see 4.3.4).

3.3.1 Metals. All metals used in the slip ring assembly shall be of corrosion resistant type or shall be suitably protected to resist corrosion during the life of the assembly. The use of dissimilar metals shall be avoided, unless suitably treated to resist galvanic corrosion (see 4.3.4).

3.3.2 Fungus resistance. All slip ring assembly materials shall be fungus non-nutrients (see 4.3.4.1.).

3.3.3 Castings. If castings are used, the casting material, type, and quality shall be as specified in the contract or order (see 4.3.4 and 6.2).

3.3.4 Seals. Seals shall be made of material compatible with MIL-PRF-6083, MIL-PRF-46170, or commercial equivalents, and water (see 4.3.4).

3.3.5 Sealed bearings. Bearings shall be thoroughly cleaned and then filled at least 80 percent (%) with grease conforming to MIL-PRF-23827, or equivalent (see 4.3.4).

3.3.6 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Operating requirements. Slip ring assemblies shall meet the operational requirements as specified in 3.4.1 through 3.4.8 (see 4.3.5).

3.4.1 Starting torque. The torque required to initiate motion of the rotor assembly shall not exceed 62.5 pound-force inch (lbf-in) (7.1 Newton-meters (N-m)) (see 4.3.5.1).

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3.4.2 Normal speed. The slip ring assembly shall operate at a continuous speed of 7.5 revolutions per minute (rpm) (see 4.3.5.2).

3.4.3 Maximum speed. The slip ring assembly shall intermittently operate at 12 rpm without sustaining damage or degrading performance (see 4.3.5.3).

3.4.4 Stopping. The rotor shall be brought from a rotational speed of 7.5 rpm to a complete stop in no more than 0.25±0.05 seconds without sustaining damage or degrading performance (see 4.3.5.2).

3.4.5 Static limit load capacity. The slip ring assembly, including the base, shall be capable of transferring each of the load combinations listed below without degrading electrical performance and without permanent structural deformation. The loads listed below shall be applied in the coordinate system as specified in figure 1 (see 4.3.5.4).

a. 1000 pounds (lb) (454 kilograms (kg)) vertically (down) applied combined with 350 lb (160 kg) horizontally applied to the slip ring assembly to produce a torque of 6000 (lbf-in) (677 N-m) about any axis in the horizontal plane.
b. 500 lb (227 kg) applied along any axis in the horizontal plane.
c. 200 lb (91 kg) vertically (up) applied along the z-axis.

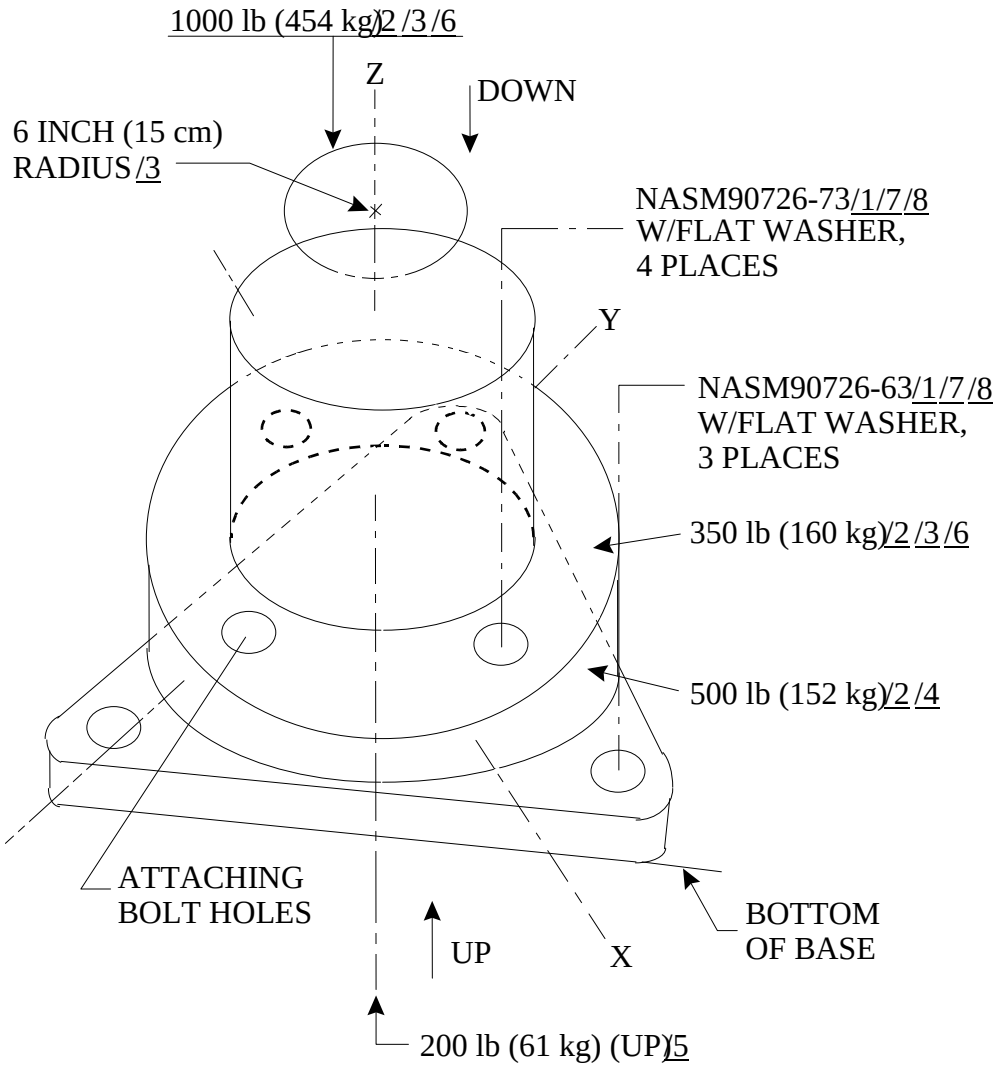
3.4.6 Electrical.

3.4.6.1 Power rings. The power rings shall function with maximum input voltage for fault free, battery only conditions and single fault conditions as specified in MIL-STD-1275. The steady state voltage shall not exceed 33 V. Class I and class II power rings shall be as specified below (see 4.3.5.5.1 and 4.3.5.5.1.1).

Class I - Class I power rings shall be designated as numbers 1 and 2 and shall transmit a current of 190 amperes (A) with surges of 500 A for up to 20 milliseconds (ms).

Class II - Class 2 power rings shall be designated as numbers 33 through 38. Rings 37 and 38 shall be high power rings and shall continuously transmit a current of 190 A, with surges of 500 A for periods up to 20 ms. Rings 33 through 36 shall be low power rings and shall continually transmit current up to 50 A, with surges of 70 A for periods up to 20 ms.

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NOTES:

1. Or commercial equivalent.
2. Apply to surface indicated, 2.34 in. (5.94 cm) above base. Load applied via attaching bolt holes.
3. Combined load case A.
4. Load case B.
5. Load case C.
6. Combined load applied 180° apart.
7. Screw length may be altered provided that suitable engagement of mating nut is maintained.
8. Torque to 180-190 lbf-in (20 to 21 N-m).
9. The Z axis is designated as vertical.
10. All axes within the x-y plane are designated as horizontal.

FIGURE 1. Coordinate system, slip ring assembly.

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3.4.6.2 Control rings. The control rings shall function with maximum input voltage for fault free, battery only conditions and single fault conditions as specified in MIL-STD-1275. The control rings shall be capable of transmitting Transistor-Transistor Logic (TTL) levels of 0.0±0.5 volts (V) direct current (dc) through 2.5±0.5 Vdc. The steady state voltage shall not exceed 33 V. Class I and class II control rings shall be as specified below (see 4.3.5.5.2 and 4.3.5.5.2.1).

Class I - Class I control rings shall be designated as numbers 3 through 17. Class I control rings shall transmit a continuous current of 7 A, with surges of 10 A allowed for not greater than 40 seconds, surges of 50 A allowed for not greater than 3 seconds, and surges of 100 A allowed for not greater than 0.8 seconds.

Class II - Class II control rings shall be designated as numbers 1 through 32. Control rings numbered 1 through 6 and 8 through 12 shall provide discrete control and transmit current as specified in 3.4.6.2.a. Control rings, numbered 14 through 19, 21 through 26, and 28 through 31, shall be capable of digital data transmission and shall transmit current at frequencies between 100 kilohertz (kHz) and 1 megahertz (MHz).

3.4.6.2.1 Capacitance. The capacitance of each control circuit, between each conductor and its shield, with the shield and the slip ring assembly housing connected to ground, shall not exceed 600 picofarads (pF) (see 4.3.5.5.2.2).

3.4.6.2.2 Noise. Measured across any two ring leads with all brush leads shorted together, control ring noise level shall not exceed 400 microvolts (μV) peak-to-peak (see 4.3.5.5.2.3).

3.4.6.3 Common requirements for power and control rings.

3.4.6.3.1 Dielectric withstanding voltage. The slip ring assembly shall be capable of withstanding 500 volts root mean square (V rms), 60 hertz (Hz), for 15 seconds between mutually insulated points without sustaining damage or degrading performance (see 4.3.5.5.3.1).

3.4.6.3.2 Insulation resistance. The insulation resistance between each two circuits and between each circuit and the case shall be not less than 100 megaohms (MΩ) at 100 Vdc at 50 to 60% relative humidity (see 4.3.5.5.3.2).

3.4.6.3.3 Resistance. The pin-to-pin resistance at 77 degrees Fahrenheit (°F) (25 degrees Celsius (°C)) through each ring shall be as specified tables IV and V (see 4.3.5.5.3.3).

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TABLE IV. Class I pin to pin resistance.

Receptacle pin, plug pin	Maximum resistance (ohms) (Ω)
A and B	0.085
C	Continuity
D through M	0.085
O	Continuity
P, R, and T	0.0063
U and V	0.085
W and X	0.0031

TABLE V. Class II pin to pin resistance.

Connector 1		Connector 2	
Receptacle pin (J1), plug pin (P1)	Maximum resistance (Ω)	Receptacle pin (J2), plug pin (P2)	Maximum resistance (Ω)
A and D	0.0043	A through F, H, J through M, P, R through Z, a, b, c, e, f, g, h	0.121
B, C, E, F	0.021	G, N, W, d, j	Continuity

3.4.7 Electromagnetic coupling. Electromagnetic coupling for Class I and II slip ring assemblies coupling shall be as specified below (see 4.3.5.6).

Class I - The current induced in a loop consisting of control rings 16 and 17, in series, by a supply current flowing in a loop consisting of power rings 1 and 2, in series, shall attenuate by not less than 70 decibels when supply currents are not less than 1 A peak and in the signal frequency of 400 to 3000 Hz. Power rings 1 and 2 shall be loaded with a resistive load of not greater than 0.2 Ω. Control rings 16 and 17 shall be loaded with a resistive load of 150 Ω on the brush side and 660 Ω of the ring side.

Class II - The current induced in a loop consisting of control rings 11 and 12, in series, by a supply current flowing in a loop consisting of power rings 37 and 38, in series, shall not attenuate more than 70 decibels. Supply currents shall be not less than 1 A peak and shall be in the signal frequency range of 400 to 3000 Hz.

3.4.8 Sheathed leads. At the point where the leads exit from the slip ring assembly, the sheathed leads shall withstand at least a 60 lb (27 kg) pull applied in any direction for at least 1 minute (see 4.3.5.7).

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3.5 Support and ownership requirements. The ring assembly shall possess the life cycle ownership characteristics as specified in 3.5.1 through 3.5.3 (see 4.3.7).

3.5.1 Safety.

3.5.1.1 Flammability. The ring assembly shall be nonflammable. Flammable impregnating and potting compounds may be used if they are enclosed in fire resistant containers (see 4.3.7.1).

3.5.1.2 Hazardous materials. The ring assembly shall not produce hazards to personnel or the environment from the use of asbestos, cadmium, radioactive materials, toxic chemicals, or other hazardous substances (see 4.3.7.1).

3.5.2 Finish. The finish for Class I and II slip ring assemblies shall be as specified below (see 4.3.7.2).

Class I - The slip ring shall be painted with semi-gloss enamel. Top coat color shall be Seafoam Green, Navy, color number 24533 of FED-STD-595 unless otherwise specified (see 6.2).
Class II - The slip ring shall be painted with an epoxy-polyamide chemical agent resistant coating conforming to MIL-PRF-22750, type I, or equivalent.

3.5.3 Identification marking. Identification and marking shall be permanent and legible and shall include as a minimum, the manufacturer's identification code (CAGE), the contract or order number, the National Stock Number (NSN) and the Part or Identifying Number (PIN) (see 4.3.7.3 and 6.5).

3.6 Operating environment requirements. The slip ring assembly shall operate under the environmental conditions as specified in 3.6.1 through 3.6.7 (see 4.3.8).

3.6.1 Water immersion (nonoperating). The slip ring assembly shall prevent leakage of water into the rotor and brush assemblies when immersed in tap water to a depth of 6 to 7 inches (in.) (15 to 18 centimeters (cm)) above the top cover (see 4.3.8.1 and 4.3.8.1.1).

3.6.2 Atmospheric pressure (nonoperating). The slip ring assembly shall withstand atmospheric pressure equivalent to an altitude of at least 40 000 feet (ft) (12 192 meters (m)) (see 4.3.8.2).

3.6.3 Sand and dust (nonoperating). The slip ring assembly shall neither be damaged nor its performance be degraded during or after exposure to windblown particles 0.000004 to 0.04 in. (0.00001016 to 0.1016 cm) in diameter from any angle, with wind speeds intermittent and variable up to 35 knots (65 kilometers per hour (km/h)) (see 4.3.8.3)

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3.6.4 Shock (nonoperating). The slip ring assembly shall withstand induced shocks of half-sine waves applied along three mutually perpendicular directions, with a peak valve of not more than 25 gravity units (g) for 11 ms (see 4.3.8.4).

3.6.5 Vibration (nonoperating). The slip ring assembly shall withstand simple harmonic motions in accordance with figure 2 applied along three mutually perpendicular directions for 3 hours along each axis (see 4.3.8.5).

3.6.6 High temperature. The slip ring assembly shall operate at a temperature of 140 °F (60 °C) and withstand a nonoperating temperature of 160 °F (71 °C) without sustaining damage (see 4.3.8.6).

3.6.7 Low temperature. The slip ring assembly shall operate at a temperature of -25 °F (-32 °C) and withstand a nonoperating temperature of -65 °F (-54 °C) without sustaining damage (see 4.3.8.7).

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.1.1).
b. Conformance inspection (see 4.1.2).

4.1.1 First article inspection. First article inspection shall be performed on preproduction (see 4.1.1.1) or initial production (see 4.1.1.2) samples when a first article sample is required (see 3.1). The inspection shall include all the verifications for requirements as specified in table VI (see 3.1 and 3.6).

4.1.1.1 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of two slip ring assemblies.

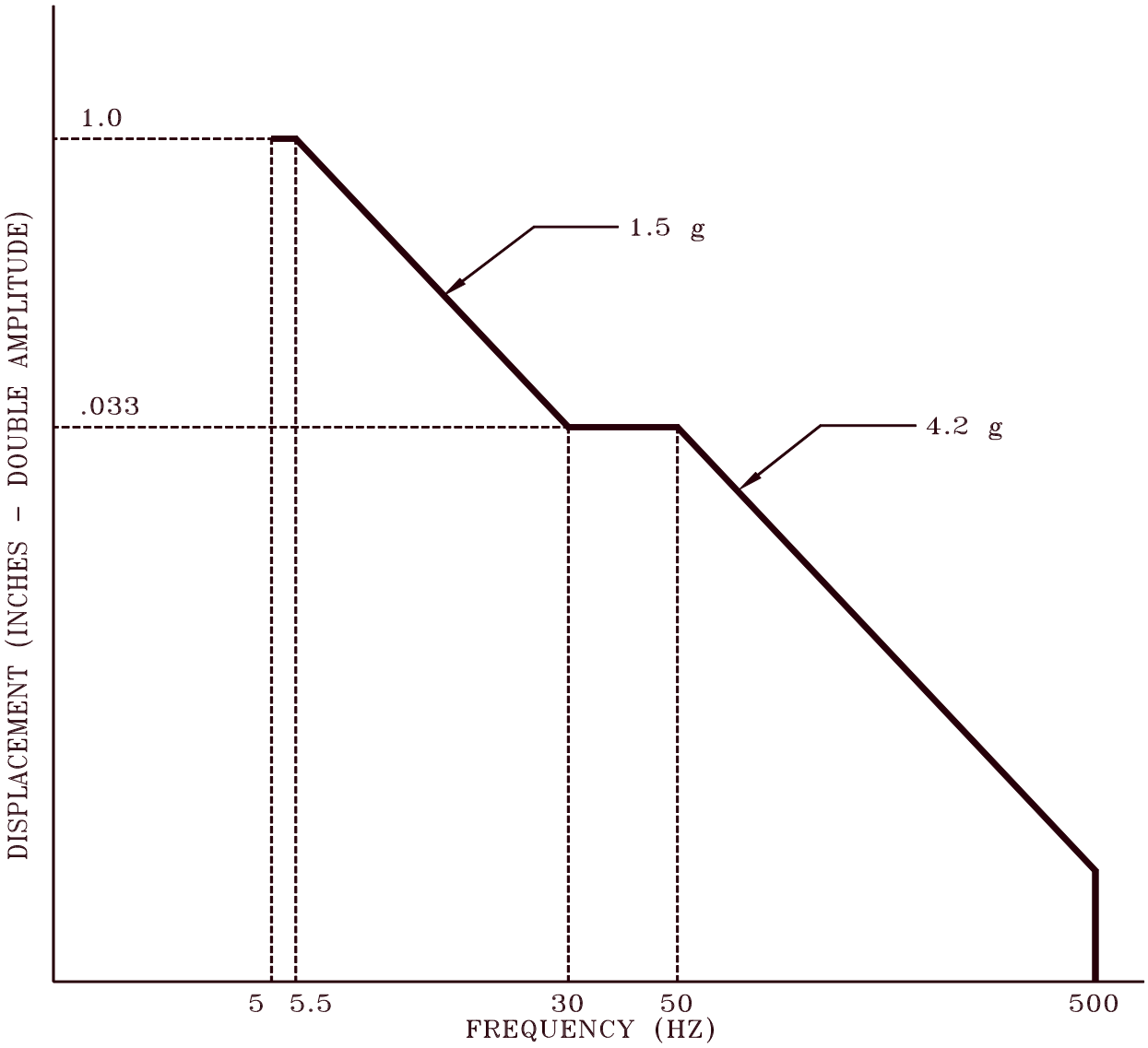
4.1.1.2 Initial production inspection. Unless otherwise specified (see 6.2), the initial production sample shall consist of two slip ring assemblies selected from the first ten units produced.

4.1.2 Conformance inspection. As specified in the contract (see 6.2 and 6.4), each slip ring assembly shall be subjected to and pass the verifications as specified in 4.3.3, 4.3.5.1 through 4.3.5.3, 4.3.5.5.2.2, 4.3.5.5.3.1 through 4.3.5.5.3.3, 4.3.7.2, 4.3.7.3, and 4.3.8.1.

4.1.2.1 Sampling. The sampling plan shall be as specified in the contract (see 6.2).

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ACCELERATION LEVELS: ±9 (PEAK)



NOTE: All curves will be extended to 2 Hz. when test item resonances below 5 Hz, are expected.

FIGURE 2. Vibration test curve.

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TABLE VI. Verification methods.

Title	Require-ments	Verifica-tion	Title	Require-ments	Verifica-tion
Design and construction	3.2	4.3.3	Capacitance	3.4.6.2.1	4.3.5.5.2.2
Housing and mounting base assembly	3.2.1	4.3.3.1	Noise	3.4.6.2.2	4.3.5.5.2.3
			Dielectric withstanding voltage	3.4.6.3.1	4.3.5.5.3.1
Brush block and terminal assembly	3.2.2	4.3.3	Resistance	3.4.6.3.3	4.3.5.5.3.3
Rotor assembly	3.2.3	4.3.3	Electromagnetic coupling	3.4.7	4.3.5.6
Interface requirements	3.2.4	4.3.3.2	Sheathed leads	3.4.8	4.3.5.7
Materials	3.3	4.3.4	**Support and ownership requirements**	3.5	4.3.7
Metals	3.3.1	4.3.4			
Fungus resistance	3.3.2	4.3.4.1	Flammability	3.5.1.1	4.3.7.1
Castings	3.3.3	4.3.4	Hazardous materials	3.5.1.2	4.3.7.1
Seals	3.3.4	4.3.4	Finish	3.5.2	4.3.7.2
Sealed bearings	3.3.5	4.3.4	Identification marking	3.5.3	4.3.7.3
Operating requirements	3.4	4.3.5	**Operating environment requirements**	3.6	4.3.8
Starting torque	3.4.1	4.3.5.1			
Normal speed	3.4.2	4.3.5.2	Water immersion	3.6.1	4.3.8.1 & 4.3.8.1.1
Maximum speed	3.4.3	4.3.5.3			
Stopping	3.4.4	4.3.5.2	Atmospheric pressure	3.6.2	4.3.8.2
Static limit load capacity	3.4.5	4.3.5.4	Sand and dust	3.6.3	4.3.8.3
Power rings	3.4.6.1	4.3.5.5.1 & 4.3.5.5.1.1	Shock	3.6.4	4.3.8.4
			Vibration	3.6.5	4.3.8.5
Control rings	3.4.6.2	4.3.5.5.2 & 4.3.5.5.2.1	High temperature	3.6.6	4.3.8.6
			Low temperature	3.6.7	4.3.8.7

4.2 Order of inspection. The inspection sequence may be in any order except that the housing and mounting base assembly test (see 4.3.3.1) and the operating environment requirements verifications (see 4.3.8) shall precede all other tests.

4.3 Verification methods. The types of verification methods included in this section are visual inspection, measurement, sample tests, full-scale demonstration tests, simulation, modeling, engineering evaluation, component properties analysis, and similarity to previously approved or previous qualified designs.

4.3.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost effective sampling procedures, to verify performance. See the contract for alternatives that replace verification required by this specification (see 6.2).

4.3.2 Inspection conditions. Unless otherwise specified (see 6.2), inspection conditions shall be as follows:

a. Ambient air temperature: 75±10 °F (24±5.5 °C)

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b. Barometric pressure: 28.54 (+2, -3) in. mercury (96.6 (+6.7, -10.1) kilopascals (kPa))
c. Relative humidity: 50±30 %

4.3.3 Design construction verification. To verify the requirements in 3.2 and 3.2.2, use one or more of the methods outlined in 4.3 to verify that the slip ring assembly is in accordance with Drawing 12265512 for class I assemblies and Drawing 12310155 for class II assemblies. Conformance shall be determined by inspection of the contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.3.3.1 Housing and mounting base assembly verification. To determine conformance to 3.2.1, mount the ring assembly in a position typical of that for which it is designed. The surface upon which the ring assembly is mounted shall extend at least 3 ft (0.91 m) beyond the equipment on all sides so that splashing may be produced by directing the water stream on that surface. Spray a water stream at 35 pounds per square inch gauge (psig) (241 kPa) with a flow of at least 10 gallons per minute (gal/min) (0.63 liters per second (L/s)). The water stream shall be sprayed from a distance of 3.0±5 ft (0.91±0.15 m) for a duration of at least 3 minutes, upon both the equipment and supporting surface from all directions on the equipment side of the surface. Any moisture entering the electrical enclosure shall be cause for rejection. A slight wetting of the mechanical surfaces between the splash seal and the electrical enclosure shall be permitted.

4.3.3.2 Interface verification. To determine conformance to 3.2.4, use one or more of the methods outlined in 4.3 to verify that the slip ring assembly is in accordance with tables II and III.

4.3.4 Materials verification. To verify the requirements in 3.3, 3.3.1, and 3.3.3 through 3.3.5, use any of the methods outlined in 4.3 to verify the following:

a. Metals shall be either inherently corrosion resistant, or suitably protected to resist the deleterious effects of corrosion.
b. If used, verify that casting are as specified in the contract or order (see 6.2).
c. Verify that seals are compatible with MIL-PRF-6083, and that sealed bearings are cleaned and at least 80% filled with grease conforming to MIL-PRF-23827, or equivalent.

Conformance shall be determined by inspection of the contractor records providing proof or certification that the materials conform to requirements in (a) through (c). Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

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4.3.4.1 Fungus (nonoperating) verification. To determine conformance to 3.3.2, in the event that the manufacturer certifies only non-nutrients have been used, the fungus test may be omitted; otherwise, the slip ring assembly shall be tested and pass the fungus test as specified in MIL-STD-810, method 508.4, or equivalent (see 4.3.1).

4.3.5 Operating requirements verification. To determine conformance to 3.4 the slip ring assembly shall be subjected to and pass the tests as specified in 4.3.5.1 through 4.3.5.7. All tests described herein shall be conducted in a suitable test fixture. The slip ring assembly shall be mounted with the mounting base horizontal as specified in figure 1.

4.3.5.1 Starting torque verification. To determine conformance to 3.4.1, apply a tangential force to the slip ring flange at a known distance from the center of rotation. The distance to the center of rotation and the applied force required to initiate motion of the rotor assembly shall be observed. The starting torque shall not exceed 62.5 lbf-in (7.1 N-m).

4.3.5.2 Normal speed verification. To determine conformance to 3.4.2 and 3.4.4, perform the applicable operational performance test as specified below.

Class I - Complete an operational cycle consisting of rotations at 7.5 rpm for 8 revolutions. The direction of rotation shall be reversed after each cycle. The slip ring assembly shall operate smoothly and continuously during direction reversal prior to the completion of 1 revolution. The rotation of the slip ring assembly shall be stopped in not greater than 0.25±0.05 seconds when changing the direction of rotation. During any 3 of the 8 revolutions in each cycle, a current of 160±16 A shall be passed through rings 1 and 2, connected in series, with a duty cycle of 10±1 second on and 30±1 second off. A current of at least 2.5 A shall be passed through rings 3 through 17 connected in series for not less than 3 revolutions in each cycle. Each current shall be inspected for loss of continuity. The slip ring assembly shall pass this test within the parameters specified without failure, degradation of performance, or damage to the unit.

Class II - Complete an operational cycle consisting of rotations at 7.5 rpm for 8 revolutions. The direction of rotation shall be reversed after each cycle. The slip ring assembly shall operate smoothly and continuously during direction reversal prior to the completion of 1 revolution. The rotation of the slip ring assembly shall be stopped in not greater than 0.25±0.05 seconds when changing the direction of rotation. During any 3 of the 8 revolutions in each cycle, a current of 160±16 A shall be passed through rings 37 and 38, connected in series, with a duty cycle of 10±1 second on and 30±1 second off. A current of 30±5 A shall be passed through rings 33 through 36 connected in series for not less than 3 revolutions in each cycle. A current of at least 2.5 A shall be passed through rings 1 through 6, 8 through 12, 14 through 19, 21 through 26, and 28 through 31, connected in series for not less than 3 revolutions in each cycle. A 1 MHz TTL square-wave signal, with a maximum current of 0.25 A, shall be passed through rings 14 through 19, 21

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through 26, and 28 through 31 for not less than 3 revolutions in each cycle. A 100 kHz TTL square-wave signal, with a maximum current of 0.25 A, shall be passed through rings 1 through 6 and 8 through 12 for not less than 3 revolutions in each cycle. Each current shall be inspected for loss of continuity. The slip ring assembly shall satisfactorily pass this test within the parameters specified without failure, degradation of performance or damage to the unit.

4.3.5.3 Maximum speed verification. To determine conformance to 3.4.3, the slip ring assembly shall complete an operational test as specified in 4.3.5.2 except that the operational cycle shall consist of rotation at 12 rpm for 8 revolutions and then reversed at 12 rpm for 8 revolutions.

4.3.5.4 Static limit load capacity verification. To determine conformance to 3.4.5, the slip ring shall be suitably mounted and then subjected to the transmitted load combinations as specified in figure 1. There shall be no degradation of electrical performance. In order to assure that the slip ring assembly is not physically damaged, verify that it is in accordance with the dimensions as specified in Drawing 12265512 for class I assemblies, and Drawing 12310155 for class II assemblies.

4.3.5.5 Electrical tests.

4.3.5.5.1 Power ring verification. To determine conformance to 3.4.6.1, the slip ring assembly shall be rotated at between 7.5 and 12 rpm while applying the maximum input voltage for fault free, battery only conditions and single fault conditions as specified in MIL-STD-1275. Steady state voltage shall not exceed 33 V. Import spike and surge voltages into the equipment as specified in MIL-STD-1275. There shall be no loss of continuity or degradation of performance.

4.3.5.5.1.1 Current test. To determine conformance to 3.4.6.1, perform the applicable verification as specified below.

Class I - Power rings designated 1 and 2 shall transmit a current of 190 A with surges of 500 A for at least 20 ms. There shall be no evidence of loss of continuity or degradation of performance.

Class II - Power rings designated as 37 and 38 shall continuously transmit a current of 190 A with surges of 500 A for periods of at least 20 ms. Rings 33 through 36 shall continuously transmit current up to 50 A with surges of 70 for periods up to 20 ms. There shall be no evidence of loss of continuity or degradation of performance.

4.3.5.5.2 Control ring verification. To determine conformance to 3.4.6.2, the slip ring assembly shall be rotated at between 7.5 and 12 rpm while applying the maximum input voltage for fault free, battery only conditions and single fault conditions as specified in MIL-STD-1275. Steady state voltage shall not exceed 33 V and the control rings shall transmit TTL levels of

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0.0±0.5 Vdc (low) and 2.5±0.5 (high). There shall be no loss of continuity or degradation of performance.

4.3.5.5.2.1 Duty cycle test. To determine conformance to 3.4.6.2, the slip ring assembly shall be rotated at 12 rpm for 8 revolutions and then reversed for 8 revolutions while carrying current as specified below. There shall be no loss of continuity or degradation of performance.

a. a. Continuous: 7 A
b. b. Transient: 10 A, 1.5 to 40 seconds
50 A, 0.15 to 3.0 seconds
100 A, 0.035 to 0.8 seconds

4.3.5.5.2.2 Capacitance test. To determine conformance to 3.4.6.2.1, the capacitance of each control circuit shall be observed between each conductor and its shield, with the shield and slip ring housing connected to ground. Capacitance shall not exceed 600 pF. Test frequency shall be between 10 and 200 kHz.

4.3.5.5.2.3 Noise test. To determine conformance to 3.4.6.2.2, all pins of the connector on the brush side shall be shorted together and the noise voltage observed across any two connector pins. A 5 milliamp (mA) current shall be passed through the rings being measured while the slip ring assembly is rotated at 8 rpm. The noise voltage shall be observed on an oscilloscope and the peak level shall not exceed 400 μV.

4.3.5.5.3 Power and control rings verifications.

4.3.5.5.3.1 Dielectric withstanding voltage test. To determine conformance to 3.4.6.3.1, apply a test potential of 500 V rms, 60 Hz, for a period of 15 seconds between all circuits and the case, and from each circuit to all others. (All subsequent testing of dielectric withstanding voltage shall be conducted at 400 V rms, 60 Hz.) There shall be no damage or performance degradation as a result of this test.

4.3.5.5.3.2 Insulation resistance test. To determine conformance to 3.4.6.3.2, an insulation resistance test shall be made between each circuit and between each circuit and the case. A test voltage of at least 100 Vdc shall be applied at 50 to 60% relative humidity. Insulation resistance shall be not less than 100 MΩ between mutually insulated points.

4.3.5.5.3.3 Resistance test. To determine conformance to 3.4.6.3.3, the resistance of each ring shall be observed from pin to pin of the input and output connectors while the slip ring assembly is rotated at a speed of 8 to 10 rpm. (Resistance measurements shall be made at intervals of 30 degrees (°) if statically determined.) The resistance shall be as specified in tables IV and V, as applicable.

4.3.5.6 Electromagnetic coupling. To determine conformance to 3.4.7, electromagnetic coupling verification shall be as specified below.

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Class I - Rings 1 and 2 shall be connected with a resistive load of at least 0.2 Ω. Rings 16 and 17 shall be loaded with a resistive load of 150 Ω on the brush side and 660 Ω on the ring side. A peak current of at least 1 A, in the frequency range of 400 to 3000 Hz, shall be applied to rings 1 and 2 with the specified load resistance. The current induced in the loop consisting of rings 16 and 17, with the specified load resistances, shall be observed. Attenuation shall be not less than 70 decibels.

Class II - Rings 11 and 12 shall be loaded with a resistive load of 150 Ω on the brush side and 660 Ω on the ring side. Rings 37 and 38 shall be loaded with a resistive load of not greater than 0.2 Ω. The current induced in a loop consisting of rings 37 and 38, shall be attenuated by not less than 70 decibels when supply currents are not less than 1 A, peak, in the frequency range of 400 to 3000 Hz.

4.3.5.7 Sheathed leads verification. To determine conformance to 3.4.8, apply at least a 60 lb (27 kg) pull for at least 1 minute to the point where the leads exit from the slip ring assembly. There shall be no evidence of physical or electrical damage.

4.3.6 Interface requirements verification. Use one or more of the methods outlined in 4.3 to determine that interface requirements are as specified in tables II and III, as applicable.

4.3.7 Support and ownership verifications. To determine conformance to 3.5, complete each verification as specified in 4.3.7.1 through 4.3.7.3.

4.3.7.1 Safety verifications. To determine conformance to 3.5.1.1 and 3.5.1.2, use one or more of the methods outlined in 4.3 to verify that the ring assembly is nonflammable, and does not produce hazards attributable to the use of asbestos, cadmium, radioactive materials, toxic chemicals, or any other hazardous substance.

4.3.7.2 Finish verification. To determine conformance to 3.5.2, visually inspect to verify that finish for class I ring assemblies is semi-gloss enamel conforming to Seafoam Green, Navy, color number 24533 of FED-STD-595 unless otherwise specified (see 6.2). Use one or more of the methods outlined in 4.3 to verify that class II ring assemblies are painted with a chemical agent resistant coating conforming to MIL-PRF-22750, or equivalent.

4.3.7.3 Identification marking verification. To determine conformance to 3.5.3, visually inspect ring assembly to verify that the CAGE, contract or order number, NSN, and PIN (see 6.5) are permanently and legibly marked on the ring assembly.

4.3.8 Operating environment requirements verifications. To determine conformance to 3.6, perform and pass each verification as specified in 4.3.8.1 through 4.3.8.7. Nonoperating environments do not require rotation or current flow through the slip ring assemblies. The unit shall not be damaged nor it’s performance degraded after environmental testing.

4.3.8.1 Water immersion (nonoperating). To determine conformance to 3.6.1, the electrical connectors on the slip ring assembly shall be capped with sealed mating connectors.

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The slip ring assembly shall be immersed in tap water to a depth of 6 to 7 in. (15 to 18 cm) above the top cover for not less than 5 minutes while bubbles which result from entrapped air on various external parts of the slip ring assembly and cable are noted and cleared where practical. While the slip ring assembly is still immersed, an internal pressure of 1±0.5 psig (7±3.5 kPa) shall be applied and held for a period of at least 1 minute while the slip ring assembly is examined for the appearance of new bubbles. There shall be no bubbles emitted from the slip ring assembly as a result of pressurization. At the completion of the test, the slip ring assembly shall be removed from the water and the pressure port and mating connector removed. The pressure port shall then be sealed with a self-sealing screw.

4.3.8.1.1 Alternate water immersion (nonoperating). If the appropriate equipment necessary for 4.3.8.1 is not available, the following test shall be performed to determine conformance to 3.6.1. The slip ring assembly shall be placed in a pressure chamber. A tube shall be connected to the slip ring assembly pressure port and routed outside the chamber to a container of water at atmospheric pressure. The end of the tube shall be covered by not more than 1 in. (2.54 cm) of water. The chamber shall be pressurized to 0.6±0.1 psig (4.1±0.7 kPa) and held for at least 3 minutes. The immersed end of the tube shall be observed for bubbles evidencing air leakage. Entrapped air in or on the tubing shall be distinguished from air leakage. There shall be no evidence of air leakage from the slip ring assembly.

4.3.8.2 Atmospheric pressure (nonoperating). To determine conformance to 3.6.2, the slip ring assembly shall be tested in accordance with MIL-STD-810, method 500.2, procedure I, or equivalent (see 4.3.1). The test pressure shall be that equivalent to an altitude of 40 000 ft (12 192 m).

4.3.8.3 Sand and dust (nonoperating). To determine conformance to 3.6.3, the slip ring assembly shall be tested in accordance with MIL-STD-810, method 510.3, procedure I, or equivalent (see 4.3.1), except that the temperature during test step 5 shall be limited to 140 °F (60 °C).

4.3.8.4 Shock (nonoperating). To determine conformance to 3.6.4, the slip ring assembly shall be tested in accordance with the requirements of MIL-STD-202, method 213B, condition J, or equivalent (see 4.3.1), except that 25 g for 11 ms shall apply.

4.3.8.5 Vibration (nonoperating). To determine conformance to 3.6.5, the slip ring assembly shall be tested in accordance with the requirements of MIL-STD-810, method 514.4, category 8, procedure I, or equivalent (see 4.3.1), with the test level as specified in figure 2. The sweep time shall be 15 minutes from 5 to 500 to 5 Hz. A sinusoidal cycling time of 30 minutes per 1000 miles (1609 kilometers (km)) or a maximum 3 hours per axis is required.

4.3.8.6 High temperature. To determine conformance to 3.6.6, the slip ring assembly shall complete 25 cycles as specified in 4.3.5.2 during high temperature testing in accordance with MIL-STD-810, method 501.3, procedures I and II, or equivalent (see 4.3.1), with the following exceptions:

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a. Operating temperature: 140 °F (60 °C)
b. Nonoperating temperature: 160 °F (71 °C)

4.3.8.7 Low temperature. To determine conformance to 3.6.7, the slip ring assembly shall complete 25 cycles as specified in 4.3.5.2 during low temperature testing in accordance with MIL-STD-810, method 502.3, procedures I an II, or equivalent (see 4.3.1), except as follows:

a. Operating temperature: -25 °F (-32 °C)
b. Nonoperating temperature: -65 °F (-54 °C)

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The slip ring assemblies covered by this specification are intended for use with a nominal 28 Vdc, power, signal, and control network supplied from the vehicle engine driven electrical system and an audio intercommunication circuit. The slip ring assemblies will provide electrical connection between circuits of a revolving turret and a fixed structure on a ground based military vehicles. Class I slip ring assemblies are intended for, but not limited to, use on the M901 Improved TOW Vehicle (ITV). Class II slip ring assemblies are intended for, but not limited to, use on the M981 Fire Support Team Vehicle (FISTV). Given that the slip ring assemblies are made exclusively for military vehicles which operate under extreme climatic conditions, this item is military unique.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. PIN and quantity of ring assemblies required (see 1.2 and 6.5).
c. If required, the specific issue of individual documents referenced (see 2.2.1, 2.2.2, and 2.3).
d. If first article samples are not required (see 3.1, 4.1.1, and 6.3).
e. Casting material, type, and quality, if used (see 3.3.3 and 4.3.4).
f. If finish and top-coat is other than specified (see 3.5.2 and 4.3.7.2).

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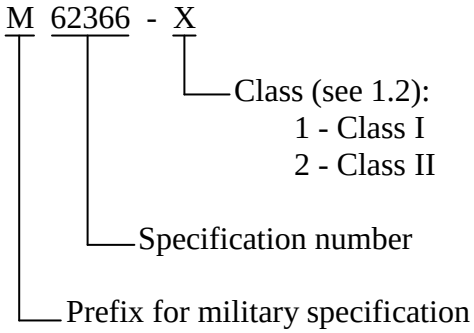
g. If preproduction inspection is required (see 4.1.1.1).
h. If initial production inspection is not required (see 4.1.1.2).
i. Specification of conformance testing and sampling (see 4.1.2 and 6.4).
j. Sampling plan (see 4.1.2.1).
k. If verification alternatives are other than specified (see see 4.3.1).
l. If inspection conditions should be other than specified (see 4.3.2).
m. Packaging requirements (see 5.1).

6.3 First article. When requiring a first article inspection, contracting documents should provide specific guidance to offerors (see 6.2). This guidance should cover whether the first article is a first article sample or a first production item and the number of test items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Pre-solicitation documents should provide Government waiver rights for samples for first article inspection to bidders offering a previously acquired or tested product. Bidders offering such products who wish to rely on such production testing must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

6.4 Conformance inspection. Affordable conformance inspection with confidence varies depending upon a number of procurement risk factors. Some of these factors include: Contractor past performance, government schedules and budget, product material and design maturity, manufacturing capital equipment and processes applied, the controlled uniformity of those processes, labor skill and training, and the uniformity of measuring processes and techniques. During the solicitation, contracting documents (see 6.2) should indicate those tests desired from table VI and their designated frequency based on a risk assessment for the procurement.

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6.5 Part or identifying number (PIN). The PIN to be used for ring assemblies acquired to this specification are created as follows:



6.6 Subject term (key word) listing.

Brush block
Control
FISTV
ITV
M901
M981
Power
Rotor

6.7 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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